

## Independent Application Answer Sheet

Rutherford Scattering Investigation - submit this completed sheet with U2 L1 Notebook Evidence.

|               |  |         |           |
|---------------|--|---------|-----------|
| Student name: |  | Date:   |           |
| TOR:          |  | Course: | Chemistry |

**Directions:** Open the linked PhET Rutherford Scattering simulation in Moodle. Observe at least 30 alpha-particle paths. Tally every observed path below. Do not estimate before observing.

| Observed path                       | Tally marks | Total |
|-------------------------------------|-------------|-------|
| Mostly straight                     |             |       |
| Slightly deflected                  |             |       |
| Strongly deflected or backscattered |             |       |

1. What do the mostly straight paths show about the structure of the atom?

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2. What do the rare large deflections or backscattering events show?

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3. State one limitation of the simulation or Rutherford model.

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**Correction record after Lesson Review:** If a response needed correction, keep the original visible and write the corrected response below.

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